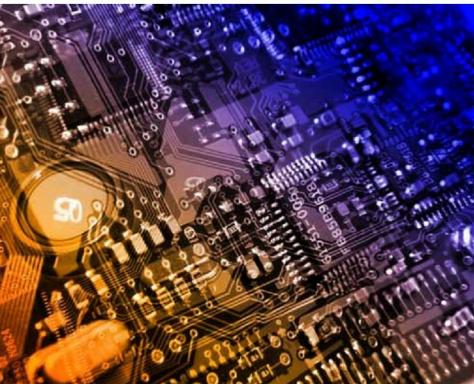


TEN TIPS

For Transition To Model-Based Design

In embedded world where delays, cost overruns and errors reflect the entire product cycle, model-based development comes to rescue. Model-based approach is being adopted by many embedded engineers and organisations for its simplicity, speed and effectiveness. Arun Mulpur and Prashant Rao from MathWorks share here the best practices for an organisation to adopt model-based development



■ SHWETA DHADIWAL BAID

Model-based approach ensures that the final product meets the end system requirements. At the same time, it increases the reusability of certain models. This approach also increases the level of interaction between different specialised work groups to identify and rectify the errors early in the development.

The transition to model-based design can be best managed by relative

study of all the aspects. Adoption and benefits of model-based design culture across the organisation are based on the 'right' combination of all these aspects. Arun Mulpur, industry marketing manager-communications, electronics and semiconductors, and Prashant Rao, technical manager, from MathWorks share here some of the best practices for an organisation for a smooth and effective transition to model-based design approach.

Tip #1: Problem analysis

It is essential to thoroughly understand the problem area before modeling any solution for it. Just like a doctor will not prescribe medications without knowing the medical history and the details of the symptoms, you need to understand relative strengths and weaknesses of the organisation and the process before adopting model-based approach. You also need to study the cost incurred due to the weaknesses and time-lines to fix those.

Once you have studied and identified the problems, prioritise the problems. Not only this, but how much time it will take to solve that problem and prove to the management the necessity of the tools and technologies is another important task.

Tip #2: Model for at least two purposes

The beauty of model-based approach is that you can design one model for multiple purposes. This is where the organisations reap the benefits of their investment in time and tools. The 'rule of two'—models produced must be used at least for two purposes—needs to be followed. For example, models used to validate requirements through simulation can also generate documentation automatically.

Although organisations can pursue single use of model, this cannot extract maximum return from the new approach. The multiple process elements